

# BMS5008: FUNDAMENTAL AND ADVANCED MULTI-OMICS RESEARCH

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## Effective Term

Semester A 2025/26

## Part I Course Overview

### Course Title

Fundamental and Advanced Multi-omics Research

### Subject Code

BMS - Biomedical Sciences

### Course Number

5008

### Academic Unit

Biomedical Sciences (BMS)

### College/School

College of Biomedicine (BD)

### Course Duration

One Semester

### Credit Units

3

### Level

P5, P6 - Postgraduate Degree

### Medium of Instruction

English

### Medium of Assessment

English

### Prerequisites

Molecular biology or genomics

### Precursors

Nil

### Equivalent Courses

Nil

### Exclusive Courses

Nil

## Part II Course Details

### Abstract

The course aims to provide lectures and practice to help students to understand the principles and applications of various -omics approaches. Topics include (1) principles of sequencing and commonly used sequencing platforms; (2) Genome sequencing transcriptome sequencing and their road applications; (3) Epigenetic modifications and regulation; (4) Transcriptional and post-transcriptional regulation; (5) Advances of proteomics and challenges; (6) Integrative data analyses and comprehensive functional genomics; (7) Gene editing, genome manipulation, emerging challenges during medical and therapeutic implementations (8) Novel inter-disciplinary and data-driven sciences. In all topics, special attention will be paid to building a fundamental understanding of omics research and applying knowledge to approach biological questions.

### Course Intended Learning Outcomes (CILOs)

| CILOs |                                                                                                                                                                                                     | Weighting (if app.) | DEC-A1 | DEC-A2 | DEC-A3 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|--------|--------|
| 1     | Describe fundamental concepts of sequencing technology and distinguish the difference between distinct sequencing platforms, and their capacity and applications in various research scenarios.     | 40                  | x      | x      |        |
| 2     | Explain the advantages and disadvantages of commonly used assays in projects that use omics approaches. Determine/apply assays to appropriately capture the desired features given certain samples. | 30                  | x      | x      | x      |
| 3     | Combine essential skills to evaluate the quality and merits of recent multi-omic research via critical thinking and simultaneously gain the ability to design or improve experiments.               | 30                  |        | x      | x      |

#### A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

#### A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

#### A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

### Learning and Teaching Activities (LTAs)

| LTAs |          | Brief Description                                                                       | CILO No. | Hours/week (if applicable) |
|------|----------|-----------------------------------------------------------------------------------------|----------|----------------------------|
| 1    | Lectures | Attend Lectures introduction and seminars to engage in the interactive learning process | 1, 2, 3  |                            |
| 2    | Tutorial | Participate in oral presentation and group discussion on a given topic for case study   | 1, 2, 3  |                            |

### Assessment Tasks / Activities (ATs)

|   | ATs                                     | CILO No. | Weighting (%) | Remarks ("- " for nil entry)                           | Allow Use of GenAI? |
|---|-----------------------------------------|----------|---------------|--------------------------------------------------------|---------------------|
| 1 | Complete assignment for presentation    | 1, 2, 3  | 40            | -                                                      | Yes                 |
| 2 | Attend lectures, tutorials and seminars | 1, 2, 3  | 20            | Attending lectures and tutorial sessions are mandatory | Yes                 |
| 3 | Final assessment                        | 1, 2, 3  | 40            | The assessment format: proposal writing or exam        | No                  |

**Continuous Assessment (%)**

100

**Minimum Continuous Assessment Passing Requirement (%)**

50

**Additional Information for ATs**

The final assessment (exam or proposal writing) covers the content of week 1-13

**Assessment Rubrics (AR)****Assessment Task**

Assignment /Attendance (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

**Criterion**

Ability to analyse and criticise the multi-omics research

**Excellent**

(A+, A, A-) Outstanding performance on all CILOs. Strong evidence of original thinking; good organization, capacity to analyse and synthesize; superior grasp of subject matter; evidence of extensive knowledge base.

**Good**

(B+, B, B-) Substantial performance on all CILOS. Evidence of grasp of subject, some evidence of critical capacity and analytic ability; reasonable understanding of issues; evidence of familiarity with literature.

**Fair**

(C+, C, C-) Satisfactory performance on the majority of CILOS possibly with a few weaknesses. Being able to profit from the course experience; understanding of the subject; ability to develop solutions to simple problems in the material.

**Marginal**

(D) Barely satisfactory performance on a number of CILOS. Sufficient familiarity with the subject matter to enable the student to progress without repeating the course.

**Failure**

(F) Unsatisfactory performance on a number of CILOS. Failure to meet specified assessment requirements, little evidence of familiarity with the subject matter; weakness in critical and analytic skills; limited or irrelevant use of literature.

**Assessment Task**

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

### **Criterion**

Ability to analyse, state and apply the principles and subject matter learnt in the course

### **Excellent**

(A+, A, A-) Outstanding performance on all CILOs. Strong evidence of original thinking; good organization, capacity to analyse and synthesize; superior grasp of subject matter; evidence of extensive knowledge base.

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### **Assessment Task**

Assignment / Attendance (for students admitted from Semester A 2022/23 to Summer Term 2024)

### **Criterion**

Ability to analyse and criticise the multi-omics research

### **Excellent**

(A+, A, A-) Outstanding performance on all CILOs. Strong evidence of original thinking; good organization, capacity to analyse and synthesize; superior grasp of subject matter; evidence of extensive knowledge base.

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## Part III Other Information

### Keyword Syllabus

Principles of sequencing,  
Genomics,  
Proteomics,  
Phylogenetic,  
Epigenetics,  
Evolution,  
Gene editing,  
Precision medicine,  
Ethics

### Reading List

#### Compulsory Readings

| Title |     |
|-------|-----|
| 1     | Nil |

#### Additional Readings

| Title |                                                            |
|-------|------------------------------------------------------------|
| 1     | Introduction to genomics, third edition, by Arthur M. LESK |